

$$(3) \quad \omega^2 = \frac{4 \pm \sqrt{7}}{3} \frac{k}{m}$$

$$\textcircled{A} \text{ 시의 대입: } -kA = (3m\omega^2 - 2k)B$$

$$-kA = (3m \cdot \frac{4 \pm \sqrt{7}}{3} \frac{k}{m} - 2k)B$$

$$-kA = (2 \pm \sqrt{7})kB$$

$$\frac{A}{B} = -2 \mp \sqrt{7}$$

$$\textcircled{i) \quad \frac{A}{B} = -2 + \sqrt{7} > 0 \Rightarrow m, 3m \text{ 같은 방향으로 운동.}$$

$$\textcircled{ii) \quad \frac{A}{B} = -2 - \sqrt{7} < 0 \Rightarrow m, 3m \text{ 반대 방향으로 운동.}$$

⑤